Emily Ing

626-297-6550 | eing@g.hmc.edu | linkedin.com/in/emilying523 | eing523.github.io/e155-portfolio

Education

Harvey Mudd College Aug. 2024 – May 2028
B.S. in Engineering, GPA: 3.7/4.0 | Dean's List *Claremont, CA*
- Coursework: Microprocessor Systems: Design & App, Digital Electronics & Computer Engineering, Continuum Mechanics, Experimental Engineering, Advanced Systems Engineering I, Applied Math for Engineering
- NCAA Division III Women's Tennis: #10 nationally, 10–2 record (2024–25)

Experience

R&D Manufacturing Engineer Contractor Aug. 2026 – Present
MiniMed - Harvey Mudd College Engineering Clinic Program *Claremont, CA*
- Design and prototype a clamping mechanism in SolidWorks to secure continuous glucose monitor sensor plates to electroplating fixtures, cutting cycle time for 192 torquing motions per lot and improving ergonomics
- Collaborate with MiniMed engineers and a student team through weekly design reviews, incorporating manufacturing and quality feedback to align the fixture with production requirements

Water Resource Control Engineer Intern May 2026 – Aug. 2026
California State Water Resources Control Board *Los Angeles, CA*
- Automated regulatory reporting for 400+ permitted facilities with R scripts, replacing a manual staff workflow
- Built an ArcGIS Pro map and Experience Builder dashboard with dynamic search, charts, and distance filtering

Undergraduate Researcher – Bioprinting Lab Jan. 2026 – Present
Harvey Mudd College - Department of Engineering *Claremont, CA*
- Engineer a UV printbed (8–10 mW/cm^2) to photopolymerize PEG-acrylate hydrogels for tissue models
- Design and prototype a multichannel print head in SolidWorks for precise, consistent bioink dispensing

Projects

FPGA Dual 7-Segment Display & Keypad | *SystemVerilog, Lattice iCE40 UltraPlus* Sep. 2026 – Sep. 2026
- Designed modular SystemVerilog to time-multiplex a dual hex display through one shared 7-segment decoder, allowing each digit to be controlled with DIP switches
- Built a counter-based scanner that drives keypad rows and reads the columns, lighting an LED on each key press

FPGA Keypad Scanner | *SystemVerilog, Lattice iCE40 UltraPlus* Sep. 2026 – Sep. 2026
- Programmed a dual 7-segment display to show the last two hex digits pressed on a 4×4 keypad scanner
- Handled switch bounce so each key press registers exactly once regardless of hold time, ignoring additional keys pressed while one is held

STM32 Digital Audio | *C, SEGGER, STM32L432KC, timers* Sep. 2026 - Sep. 2026
- Wrote TIM6/TIM7 drivers in C for note frequency and duration
- Derived prescaler/auto-reload values for minimum and maximum note frequencies to play tunes through an LM386 amplifier with every note within 1% of target pitch

STM32 Motor Speed & Direction Reading | *C, SEGGER, STM32L432KC, interrupts* Oct. 2026 – Oct. 2026
- Measured motor speed and direction from a quadrature encoder by using interrupts on both sensors

Autonomous Surface Vehicle (ASV) | *Analog circuit design, sensor integration* Mar. 2026 – Apr. 2026
- Built and deployed an ASV to measure ocean pollution across 0–2m depths on a $50 budget
- Designed and waterproofed analog signal conditioning circuits for pressure, turbidity, and temperature sensors

Autonomous Underwater Robot | *LabVIEW, PCB design, control systems* Aug. 2025 – Nov. 2025
- Soldered PCB circuitry to integrate pressure sensor and thermistor, validating signal integrity on an oscilloscope
- Programmed a LabVIEW proportional control system for autonomous depth tracking and step response analysis

Skills

Languages: C, SystemVerilog, Python, MATLAB, R, Java, JavaScript
Software: Lattice Radiant, SEGGER, Quartus Prime, Questa, LabVIEW, SolidWorks, COMSOL, ArcGIS Pro
Hardware & Lab: STM32, Lattice iCE40 FPGA, Oscilloscopes, PCB design/soldering, Analog Signal Conditioning
Fabrication: Mill, Lathe, Laser Cutter, 3D Printer